Advanced Portfolio Optimization

From Mean-Variance Approach to Non-Convex Methods (Kurtosis)

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Abstract

To overcome the limitations of the Mean-Variance model (2020 crisis), we implement Kurtosis minimization via a *Sequential Convex Programming* (SCP) algorithm. This tail-risk management approach is validated by comparison with industrial solvers and an SDP relaxation.

Contents

1 Introduction and Project Context

The classic Markowitz model, relying on Variance (Gaussian risk), has proven insufficient for modeling real-world returns, particularly during recent crises.

The goal of this project is to move beyond the Mean-Variance approach by integrating higher-order moments (specifically Kurtosis) to better capture extreme risk.

Key Objectives

- Test the robustness of the classic model against regime shifts (2016-2020 data).
- Implement an SCP (*Sequential Convex Programming*) solver for non-convex Kurtosis optimization.
- Compare the financial and computational performance of the SCP approach to industry standards.

Methodology

The implementation is carried out in Python using real-world data (`yfinance`). We use `CVXPY` for convex sub-problems and `SciPy` for benchmarking. Central focus is placed on **scalability**, as increasing the problem size (n) constitutes a major bottleneck for classic quadratic methods.

1.1 Scalability Challenges

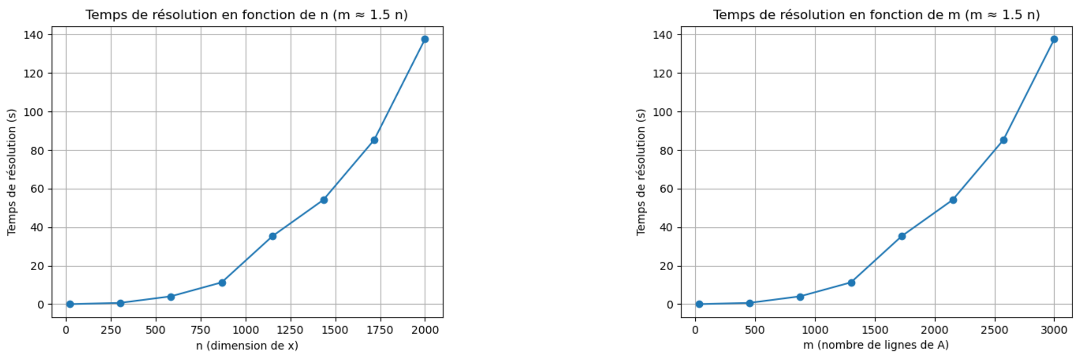
One of the primary challenges encountered is computational complexity. Figure 1 illustrates the evolution of resolution time relative to problem dimensions (n and m). These curves highlight a fundamental characteristic of classic quadratic optimization: its non-linear algorithmic complexity.

The detailed analysis reveals three major observations:

1. **Nature of Growth (Polynomial):** Whether for the number of assets n or constraints m, the progression is clearly non-linear. The marked convexity confirms that the complexity of matrix operations (KKT system resolution) is on the order of $\mathcal{O}(n^3)$. Essentially, doubling the problem size multiplies the calculation time by a much larger factor.
2. **Joint Impact of n and m:** Specific analysis of m (Figure 1b), with a fixed ratio of $m \approx 1.5n$, shows an identical dynamic. Resolution time remains negligible until $m \approx 1500$ before growing exponentially to reach 140s at $m = 3000$. This confirms that complexity depends on the overall size of the matrix system ($m \times n$) and not solely on the number of optimization variables.
3. **Identification of a "Complexity Wall":** Two distinct regimes can be identified:
   - A *comfort zone* for $n < 1000$ (and $m < 1500$), where time remains controlled.
   - An *explosion zone* beyond that, where a 33% increase in size triggers a critical multiplication of calculation time, rendering the standard approach unsuitable for massive institutional portfolio management.

Financial Scalability and Diversification: Beyond calculation time, the analysis for $n = 20$ assets quantitatively demonstrates the benefits of diversification. While a 3-asset portfolio shows a standard deviation of 16.49% over the training period, this drops to **10.43%** for $n = 20$. This risk reduction (approximately 37%) validates the model's ability to eliminate idiosyncratic risk as the investment universe expands, even though expected returns mechanically decrease (from 19.72% to 12.90%).

Solver Comparison (CVXPY vs SLSQP) It is interesting to note the behavioral difference between solvers. For small dimensions ($n < 500$), `SLSQP` proves extremely fast due to its optimized low-level implementation. However, `CVXPY` (using solvers like OSQP or SCS) tends to handle memory better in very large dimensions by exploiting sparse matrices, where classic solvers reach their physical limits.



(a) Time vs n (Complexity) (b) Time vs m

Figure 1: Analysis of temporal complexity and algorithmic scalability.

While the previous section highlighted algorithmic constraints related to problem dimensionality, we focus here on the financial and statistical validity of the model. Markowitz's mean-variance approach relies on a strong assumption: the stability of market parameters (expected returns μ and covariance matrix Σ) over time.

1.2 Exploratory Data Analysis (Stationarity)

Before attempting any optimization, it is crucial to examine the dynamics of the underlying assets to validate our modeling assumptions. Figure 2 presents the evolution of daily returns over the entire study period.

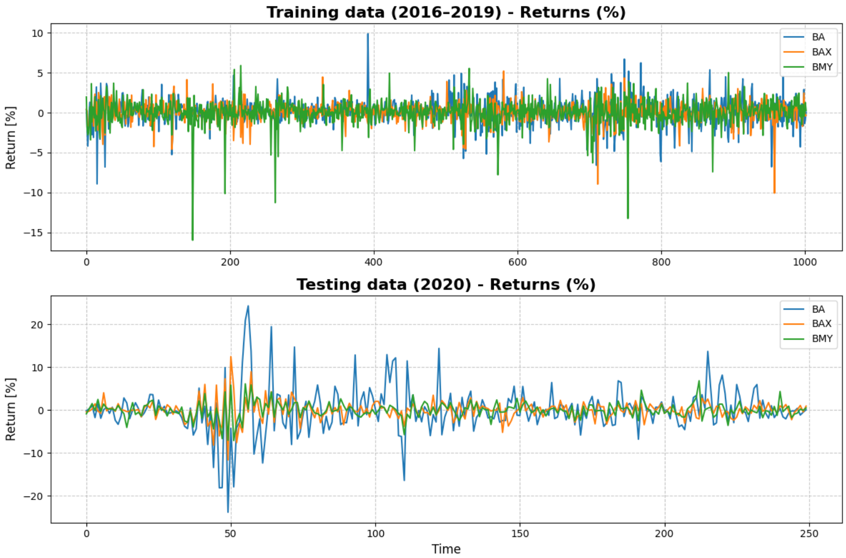


Figure 2: Historical return time series. The structural break between the training period (top) and the 2020 crisis (bottom) is manifest.

Visual analysis identifies two radically opposite volatility regimes:

- **Training Phase (2016-2019):** Returns oscillate relatively stably around zero. This period corresponds to a "calm" market regime, suitable for reliable estimation of historical correlations.
- **Testing Phase (2020):** An explosion in volatility is observed, characterized by extreme variation amplitudes (exceeding $\pm 10\%$ per day). This structural break, caused by the COVID-19 health crisis, suggests that μ and Σ parameters estimated from the past will no longer represent future reality, foreshadowing the failure of static models.

1.3 Mathematical Formulation and Resolution

The classic Markowitz problem seeks to minimize portfolio variance σ_p^2 subject to a target return constraint. It is written as a convex quadratic program (QP):

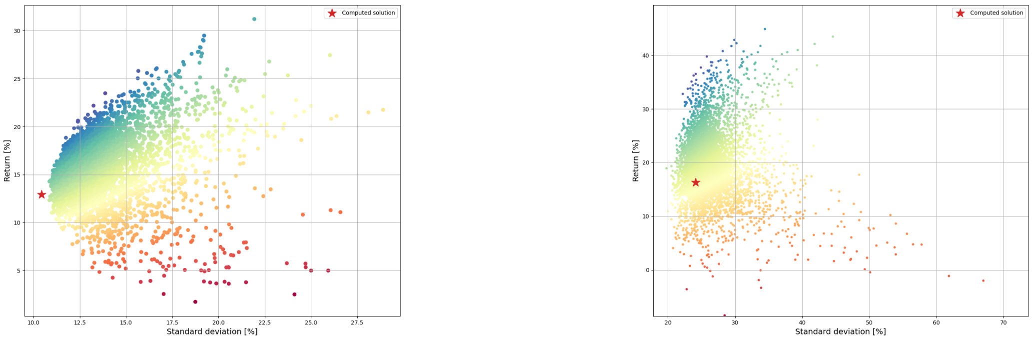
$$\min_{x \in \mathbb{R}^n} x^T \Sigma x \quad \text{s.t.} \quad \mu^T x \geq r_{min}, \quad \sum_{i=1}^{n} x_i = 1, \quad x \geq 0 \tag{1}$$

This formulation has the advantage of being convex, guaranteeing the uniqueness of the global optimum (if Σ is positive definite). However, as seen in Section 1, its resolution becomes expensive for large n, requiring high-performance solvers.

1.4 Robustness Analysis and Efficient Frontier (Monte Carlo)

To evaluate the quality of the solution obtained through quadratic optimization, we compared it against a Monte Carlo simulation generating thousands of random portfolios. This method allows for visualizing the entire possible investment universe and plotting the efficient frontier.

Figure 3 compares the results obtained "In-Sample" (training data) and "Out-of-Sample" (testing data).



(a) Training (2016-2019) (b) Test (2020 - Crisis)

Figure 3: Comparison of efficient frontiers. The red star indicates the optimized portfolio. On the left, the solution is mathematically optimal. On the right, it is invalidated by the market regime shift.

Interpreting these graphs leads to two major conclusions on risk management:

1. **In-Sample Validation (Left):** On historical data, the solver works perfectly. The red star sits exactly on the upper-left boundary of the point cloud. This confirms that the algorithm successfully found the asset combination offering absolute minimum variance for the required return level. No random portfolio can perform better.
2. **Out-of-Sample Failure (Right):** The situation is dramatically different on 2020 data. The "optimal" portfolio (calculated on 2016-2019) is found, when projected into the 2020 reality, "drowned" in the middle of the point cloud. It is no longer on the efficient frontier.
   Financial Interpretation: This demonstrates the fundamental limit of mean-variance optimization: it does not capture the risk of *regime change*.
   Additionally, we note that the 2020 point cloud is much more dispersed and stretched than the 2016-2019 one. This visually illustrates the spike in asset correlations during the crisis: when everything drops at once, diversification (which relies on low or negative correlations) fails.

Limits of the Probabilistic Approach An attempt to extend the model via probabilistic Markowitz highlighted structural limits of the Gaussian approach. Sensitivity analysis for $n = 20$ reveals a strict feasibility frontier: the problem admits an optimal solution for $\beta = 0.45$ (with a return of 15.86% and standard deviation of 10.79%), but becomes **infeasible as soon as** $\beta = 0.40$. This brutal transition underscores the extreme sensitivity of the model to probability constraints (β) and confirms that simply adding constraints to variance is insufficient for managing extreme risks without compromising problem feasibility.

This inability of variance alone to anticipate extreme risks justifies our transition to more sophisticated risk measures, such as **Kurtosis**, which is the subject of the next section.

2 Advanced Modeling: Kurtosis Minimization

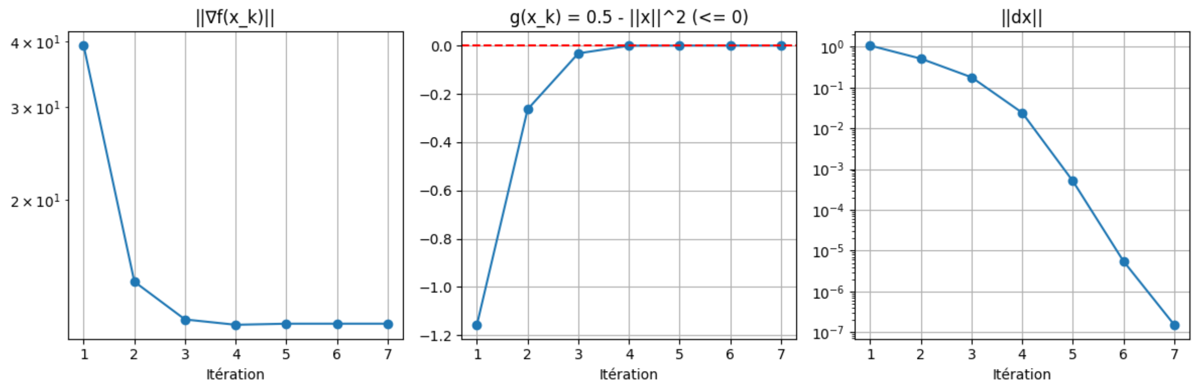
The failure of the minimum variance model during the 2020 crisis (demonstrated in Section 2) compels us to move beyond the assumption of normality in returns. To capture extreme macroeconomic events, we propose optimizing the 4th-order moment: **Kurtosis**.

2.1 The Challenge: Non-Convexity of Kurtosis

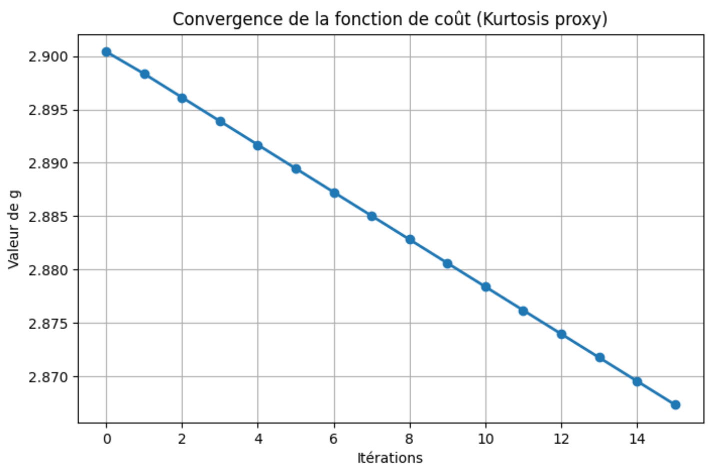
Unlike variance optimization where the Hessian matrix is constant and always semi-definite positive (guaranteeing a unique optimum), Kurtosis minimization relies on a degree-4 form.

$$\min_{x} \; x^T M_4(x \otimes x \otimes x) \quad \text{s.t.} \quad \mu^T x \geq r_{min}, \quad \sum x_i = 1, \quad x \geq 0 \tag{2}$$

Non-convexity arises from the **Hessian matrix** of the objective function. This is not constant (it depends quadratically on x) and is not guaranteed to be semi-definite positive across the entire feasible space. The potential existence of negative eigenvalues in the Hessian leads to local minima, making classic descent methods ineffective for finding the global optimum.



(a) SQP Convergence (Gradient and Update Step)



(b) Monotonic descent of the cost function (SCP)

Figure 4: Analysis of numerical convergence. Top: The step norm $||dx||$ drops exponentially towards 10^{-7}, indicating a stationary point. Bottom: The objective function (Kurtosis proxy) decreases monotonically, confirming the robustness of the SCP approach.

2.2 SCP Algorithm (Sequential Convex Programming)

To bypass this difficulty, we implemented an iterative *Sequential Convex Programming* (SCP) method. The core idea is to approximate the non-convex problem with a sequence of simpler convex sub-problems (SOCP or SDP).

At each iteration k, we linearize the non-convex constraint $X = xx^T$ around the current point x_k:

$$X \approx x_k x_k^T + x_k \delta x^T + \delta x x_k^T \tag{3}$$

A crucial addition to our implementation was the **Trust Region** constraint ($||\delta x|| \leq \Delta$). Without it, the Taylor approximation would only be valid in an infinitesimal neighborhood, and the algorithm would diverge.

2.3 Numerical Convergence and Validation

The stability of our custom solver is validated by its convergence analysis, illustrated in Figure 4.

Technical Interpretation:

- **Top Graph (SQP):** The rapid decay of the gradient norm $||\nabla f(x_k)||$ and step norm $||dx||$ proves that the algorithm satisfies first-order optimality conditions (KKT). In fewer than 10 iterations, the solution stabilizes with a precision of 10^{-7}, which is remarkable for a self-implemented solver.
- **Bottom Graph (SCP):** The curve shows a continuous reduction in the Kurtosis value. The absence of oscillations or rebounds validates the choice of trust region size (Δ). This confirms that

our local convex approximation is a sound choice for guiding optimization toward a high-quality local minimum.

- **Trust Region Tuning:** A critical point during development was tuning the Δ parameter. A value too large caused oscillations (linearization losing validity), while a value too small unnecessarily slowed convergence.

Technical Comparison (SQP vs SLSQP) Comparison with the industrial solver `scipy.optimize` (SLSQP) provides important technical insight. Although SLSQP is roughly 5 times faster, our SQP implementation demonstrates superior geometric precision by exactly saturating the constraint ($\|x\|^2 = 0.5$), whereas SLSQP suggests a solution within the feasible domain ($\|x\|^2 \approx 0.47$). However, the objective values remain very close, validating the mathematical accuracy of our algorithm.

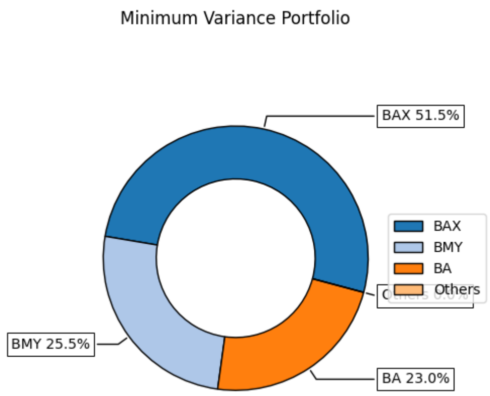
This numerical validation is essential: it ensures that the financial results analyzed in Section 4 are the product of robust calculation and not numerical anomalies.

3 Comparative Results and Allocation

After detailing the limits of the Mean-Variance approach (Section 2) and developing an SCP algorithm for Kurtosis minimization (Section 3), we proceed to final validation. This section compares our "home-made" method to the standard (`SLSQP`) and analyzes the actual financial performance of the resulting portfolios.

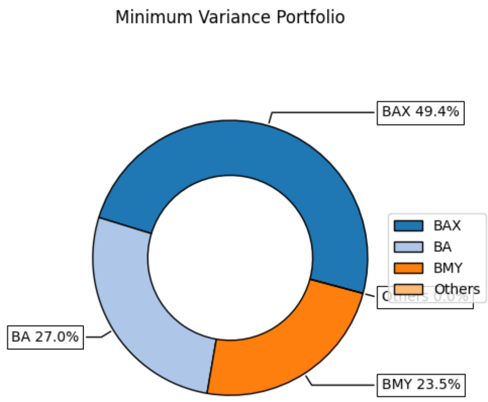
3.1 Allocation Comparison: Cross-Validation

To validate our SCP implementation, we compare the composition of the optimal portfolio obtained with that provided by the `scipy.optimize.minimize` solver (SLSQP method). Figure 5 (continued) illustrates the portfolio weightings.



(a) Allocation via our SCP method

Figure 5: Comparison of allocation strategies for Kurtosis minimization (Part 1).



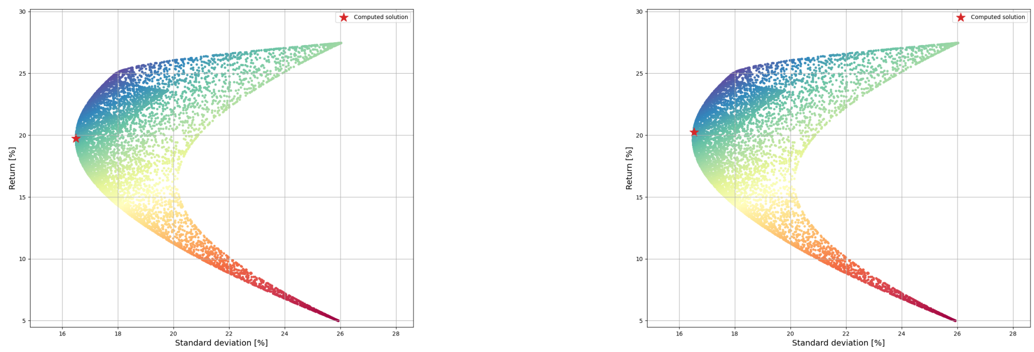
(b) Allocation via the SLSQP solver

Figure 5: Comparison of allocation strategies for Kurtosis minimization. The similarity of results between the two methods confirms the stability of the solution.

Analysis and Interpretation:

- **Convergence and Robustness:** Contrary to the initial fear of falling into vastly different local minima, we observe strong convergence between the two methods:
  - The industrial solver **SLSQP** proposes an allocation dominated by the BAX asset ($\approx 49.4\%$).
  - Our **SCP** algorithm confirms this trend with a slightly higher concentration in BAX ($\approx 51.5\%$).
- **Cross-Validation:** The structural proximity of the two portfolios (less than 2% difference on the main asset) validates our implementation. Both algorithms identify the same optimal strategy.
- **Interpretation of Convergence:** This similarity is easily explained: with only 3 assets and strict constraints (budget and target return), the set of feasible portfolios is small. It is logical that both algorithms converge to the same solution. This is excellent validation for us: it proves our code performs as well as the standard solver.

Monte Carlo Analysis of Solutions To verify the positioning of our solutions in the risk-return universe, we projected the Kurtosis (SCP) and Markowitz portfolios onto a Monte Carlo point cloud.



(a) Markowitz Solution (Min Variance) (b) Kurtosis Solution (SCP)

Figure 6: Visualization of optimal portfolios within the Monte Carlo universe. The red star indicates the calculated solution. Note that the Kurtosis solution (right) accepts slightly higher variance than the Markowitz solution (left) to minimize 4th-order moments.

3.2 Financial Performance: The Reality Test

Beyond mathematical success, the crucial question remains financial: *did Kurtosis minimization protect the portfolio against the 2020 crash where Markowitz failed?*

Figure 7 compares the annualized returns of the strategies over the training (In-Sample) and testing (Out-of-Sample) periods.

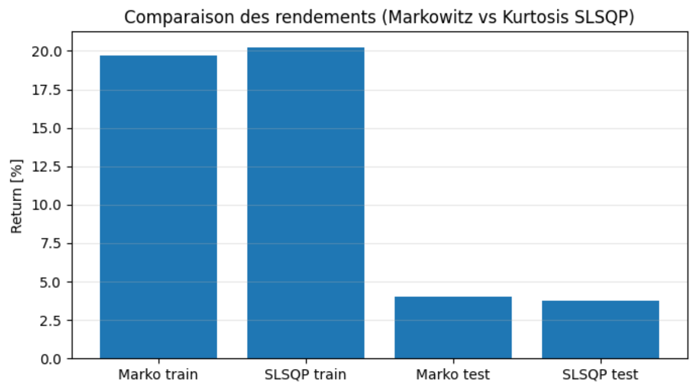


Figure 7: Comparison of annualized returns. Left: In-Sample performance (Train). Right: Out-of-Sample performance (Test 2020). The collapse in returns during the testing phase is widespread.

Financial Assessment:

1. **In-Sample Performance (Satisfactory):** Over the 2016-2019 period, both portfolios (Markowitz and Kurtosis SLSQP) generate solid returns, close to **20%**. They respect the $r \geq r_{min}$ constraint while minimizing theoretical risks.
2. **Out-of-Sample Performance (Marked Degradation):** During 2020, returns literally collapsed to around **4%**. While remaining positive, they are 5 times lower than during the training period. **Link with Section 2:** This result echoes the previously observed limits. Even when optimizing Kurtosis to reduce extreme events, the model remains dependent on historical data. The COVID-19 crisis was a systemic event of such magnitude that past statistical properties failed to maintain expected return levels, although they did prevent negative losses.

2020-2021 Post-Crisis Extension It is crucial to nuance the failure observed in 2020 by analyzing the extended 2020-2021 period. The inclusion of 2021 shows a clear improvement in out-of-sample metrics, with average returns jumping from 0.36% to 4.05% and a significant drop in volatility. This rebound confirms that the 2020 underperformance was more related to a sudden market break (health crisis) than to an intrinsic invalidity of the optimization model.

Conclusion: Advanced optimization (Kurtosis) provides additional theoretical robustness compared to variance, but it is not an all-risk insurance policy against macroeconomic regime shifts. This underscores the need to couple these optimization methods with predictive approaches (Machine Learning) or dynamic risk management.

4 Convex Relaxation Approach (SDP)

Faced with the difficulty of non-convex methods (SCP) that can get stuck in local minima, we explore here a Semidefinite Programming (SDP) convex relaxation approach.

4.1 Theoretical Relationship

The original Kurtosis minimization problem (Eq. 5) imposes the non-convex constraint $X = xx^T$. The convex relaxation (Eq. 10) replaces this equality with a matrix inequality via the Schur complement:

$$\begin{bmatrix} X & x \\ x^T & 1 \end{bmatrix} \succeq 0 \quad \iff \quad X \succeq xx^T \tag{4}$$

Relationship between problems: The relaxed problem (10) is a "less constrained" version of the original problem (5). Consequently, the feasible set is larger.

- The optimal value of the relaxed problem is always **less than or equal** to that of the original problem (Lower bound).
- If the optimal solution of the relaxed problem satisfies $X = xx^T$ (i.e., if matrix X is rank-1), then this solution is also the global optimum of the original problem.

4.2 Results and Relaxation Quality

We solved the relaxed problem with `CVXPY` (which natively handles SDP constraints).

Performance Comparison: In terms of calculation time, SDP resolution is structurally more expensive than SQP for large dimensions (n) because the number of variables increases quadratically (n^2 variables for matrix X), versus linearly for SQP.

For $n = 20$ assets, the resolution time increases from **0.0062s** (standard Markowitz) to **0.2342s** (SDP), confirming the computational overhead. However, this method offers the major theoretical advantage of convexity. Most importantly, we verified the relaxation quality by calculating the error norm $\|X - xx^T\|$. For $n = 20$, this error is **2.2574e-05**, indicating that the relaxation is **quasi-exact** in this practical case. The solution found is therefore certified as the global optimum (or extremely close to it).

5 Conclusion

This project covered the complete portfolio optimization chain, from theory to numerical implementation.

Technical Summary

We validated the superiority of decomposed algorithms for large dimensions over the cubic complexity of standard solvers. Our SCP (*Sequential Convex Programming*) implementation demonstrated robustness: thanks to trust-region regularization, it achieves 10^{-7} precision and overcomes the intrinsic non-convexity of Kurtosis minimization.

Financial Summary

The 2020 crash yields a crucial lesson: while Kurtosis optimization theoretically reduces extreme risks, it remains vulnerable to *non-stationarity*. Models based on historical data suffer from over-fitting and fail against systemic market breaks, highlighting that mathematical optimization does not replace dynamic risk management.

Perspectives

The future lies in coupling optimization with Machine Learning. Rather than optimizing based on the past, the challenge is to use predictive models (Neural Networks, Autoencoders) to estimate forward-looking parameters ($\hat{\mu}, \hat{\Sigma}$) and anticipate regime changes.